

A .
(Square $n \times n$)

$$\underline{A} \underline{x} = \lambda \underline{x} \quad (\text{non-zero } \underline{x}, \text{ numbers } \lambda).$$

$$\underline{A} \underline{x} - \lambda \underline{I}_n \underline{x} = \underline{0}$$
$$(\underline{A} - \lambda \underline{I}_n) \underline{x} = \underline{0}$$

A has free cols iff $\det(A) = 0$.

If A has free cols; say $\underline{c}_j = \sum_{i < j} \alpha_i \underline{c}_i$

$$\det \begin{bmatrix} | & & | & | & & | \\ \underline{c}_1 & \dots & \underline{c}_{j-1} & \underline{c}_j & \dots & \underline{c}_n \\ | & & | & | & & | \end{bmatrix} = \det \begin{bmatrix} | & & | & & & | \\ \underline{c}_1 & \dots & \underline{c}_{j-1} & \alpha_1 \underline{c}_1 + \alpha_2 \underline{c}_2 + \dots + \alpha_{j-1} \underline{c}_{j-1} & & \underline{c}_n \\ | & & | & & & | \end{bmatrix}$$
$$= \sum_{i < j} \alpha_i \det \begin{bmatrix} | & & | & | & & | \\ \underline{c}_1 & \dots & \underline{c}_{j-1} & \underline{c}_i & \underline{c}_{j+1} & \dots & \underline{c}_n \\ | & & | & | & & | \end{bmatrix} = 0.$$

If $\det(A) = 0$ then it has free cols.

OR If A has no free cols, then $\det(A) \neq 0$.

$\det(A) = \pm$ product of pivots $\neq 0$.

depending on how many row exchanges in elimination.

Check $\det(A - \lambda I) = 0$.

$$A = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} 1-\lambda & 0 \\ -1 & 1-\lambda \end{pmatrix}$$

$$\lambda I = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

$\det(A - \lambda I) = ?$ use linearity properties.

First row $(1-\lambda) \begin{pmatrix} 1 & 0 \end{pmatrix} + 0 \begin{pmatrix} 0 & 1 \end{pmatrix}$

$$\det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & [1 \ 0] \\ -1 & (1-\lambda) \end{pmatrix} + \det \begin{pmatrix} 0 & [0 \ 1] \\ -1 & 1-\lambda \end{pmatrix}$$

$$= (1-\lambda) \det \begin{pmatrix} 1 & 0 \\ -1 & 1-\lambda \end{pmatrix} + 0 \cancel{\det \begin{pmatrix} 0 & 1 \\ -1 & 1-\lambda \end{pmatrix}}$$

Second row: $-1 [1 \ 0] + (1-\lambda)[0 \ 1]$

$$(1-\lambda) \det \begin{bmatrix} 1 & 0 \\ -1 [1 \ 0] + (1-\lambda)[0 \ 1] \end{bmatrix} = (1-\lambda) \det \begin{bmatrix} 1 & 0 \\ -1 [1 \ 0] \end{bmatrix} + (1-\lambda) \det \begin{bmatrix} 1 & 0 \\ (1-\lambda)[0 \ 1] \end{bmatrix}$$

$$= (1-\lambda)(-1) \det \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} + (1-\lambda)^2 \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= (1-\lambda)^2$$

$$A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \quad \det(A - \lambda I) = (1-\lambda)^2$$

roots of $\det(A - \lambda I) = 0$ are 1, 1.

$$A - 1I = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} - 1 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix}$$

$$\text{null}(A - I) = \left\{ \alpha \begin{pmatrix} 0 \\ 1 \end{pmatrix} : \alpha \in \mathbb{R} \right\}$$

$$\text{rref}(A - I) = \text{rref} \begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} \underline{C2} = 0 \underline{C1} \\ \underline{C2} - 0 \underline{C1} = \underline{0} \end{array}$$

characteristic polynomial.

Roots of $\det(A - \lambda I) = 0$: Eigenvalues.

If a root is repeated k times in $\det(A - \lambda I) = 0$, we say that eigenvalue has algebraic multiplicity k .

Square Symmetric real A :

$$A^T = A \quad \text{eg: } A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

$$A - \lambda I = \begin{pmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{pmatrix}.$$

$$\det(A - \lambda I) = (1-\lambda)^2 - 1 = \lambda^2 - 2\lambda + 1 - 1 = \lambda^2 - 2\lambda.$$

Roots: 0, 2.

$$\text{null}(A - 0I) = \text{null}(A) = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} : \alpha \in \mathbb{R} \right\}.$$

$$\text{rref}(A) = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \quad \begin{array}{l} \underline{c_2} = -\underline{c_1} \\ \underline{c_2} + \underline{c_1} = \underline{0}. \end{array}$$

$$\text{null}(A - 2I) = \text{null}\left[\begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix}\right] = \left\{ \alpha \begin{pmatrix} 1 \\ -1 \end{pmatrix} : \alpha \in \mathbb{R} \right\}.$$